

SAT Math Mastery · Week 6B**Center, Spread & Plots****Time** 150 minutes: 110 min solving + 40 min self-check and error log**Topics** mean, median, mode, range · standard deviation · adding, removing and changing a value · weighted means · frequency tables · box plots, dot plots, histograms · comparing spreads without computing**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Compute mean, median, mode and range from a list, a frequency table or a dot plot.
- S2.** Use the total: $\text{sum} = n \times \text{mean}$; find a missing value that gives a target mean.
- S3.** Predict the effect of adding, removing or changing one value on the mean and on the median position.
- S4.** Combine groups with a weighted mean; recover a group size from the combined mean.
- S5.** Read the median from running totals (frequency tables, dot plots, histograms with bins).
- S6.** Read a box plot: five-number summary, IQR, range; know what a box plot cannot show (the mean).
- S7.** Compare standard deviations by distance from the mean, without computing.
- S8.** Apply $y = ax + b$ to a data set: effect on mean, median, range, IQR and standard deviation.
- S9.** Bound the possible mean or median from a histogram or a partly known data set.

2 · REQUIRED FACTS

- Mean = $\frac{\text{sum}}{n}$. Median = middle value of the ordered list (average of the two middle values when n is even). Mode = most frequent value. Range = $\text{max} - \text{min}$.
- The median is found by position, so changing an extreme value does not move it; the mean does move.
- Adding a value equal to the mean leaves the mean unchanged and *reduces* the standard deviation.
- Standard deviation measures typical distance from the mean: values farther from the mean raise it, values closer lower it.
- Adding a constant to every value shifts mean and median but leaves range, IQR and SD unchanged.
- A box plot shows min, Q_1 , median, Q_3 , max; each of the four sections holds about 25% of the data. It does not show the mean.

3 · REQUIRED FORMULAS

$$\bar{x} = \frac{\sum x}{n}, \quad \bar{x}_{\text{new}} = \frac{n\bar{x} + v}{n+1}, \quad \bar{x}_{\text{combined}} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}, \quad \text{median position} = \frac{n+1}{2}$$

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}, \quad \text{IQR} = Q_3 - Q_1, \quad \text{mean from a table} = \frac{\sum fx}{\sum f}$$

$$y = ax + b: \quad \bar{y} = a\bar{x} + b, \quad \text{median}_y = a \text{median}_x + b, \quad \sigma_y = |a|\sigma_x, \quad \text{range}_y = |a|\text{range}_x$$

4 · MINI THEORY SHEET

Which measure answers which question.

Question	Tool
a missing value for a target mean	target sum – known sum
effect of one new value v	new mean = $\frac{n\bar{x}+v}{n+1}$; the median moves at most one position
two groups together	weighted mean (weights are the group sizes)
median from a frequency table	running total until it passes $\frac{n+1}{2}$ (or the two middle positions)
which set is more spread out	compare distances from the mean, not the range
a histogram's mean	least: every value at the left edge; greatest: right edge

Box plot. The ends of the box are Q_1 and Q_3 , the line inside is the median, the whiskers reach min and max. Range = max – min; IQR = box length.

Dot plot and histogram. Each dot is one value, so count dots in order to reach the middle position. In a histogram, bar heights are frequencies; the total is the sum of the heights. The median lies in the first bar whose running total reaches the middle position(s).

Transformations. Multiplying all values by a scales the spread by $|a|$; adding b shifts the center by b and does not change the spread.

Speed shortcuts.

Situation	Shortcut
add a value to reach a target mean	needed value = target mean + (target mean – old mean) $\times n$
replace one value u by w	mean changes by $\frac{w-u}{n}$; the median is unchanged unless the middle moves
add c to every value	mean and median shift by c ; range, IQR, SD unchanged
two groups, combined mean	the combined mean sits closer to the larger group
SD without computing	mostly-far-from-mean beats a long thin tail of near values
histogram mean bounds	use left edges for the least, right edges for the greatest
box plot counts	each quarter holds 25% of the values

5 · GUIDED EXAMPLES

Example 1 — Total, new mean and median

The mean of 6 values is 14. A seventh value, 28, is added. Find the new mean. Then find the median of 2, 5, 5, 9, 12, 20.

Total of the six: $6(14) = 84$. New mean = $\frac{84+28}{7} = \frac{112}{7} = 16$.

Six values (even n): median is the average of the 3rd and 4th: $\frac{5+9}{2} = 7$.

Example 2 — Frequency table: mean and median

Scores 1, 2, 3, 4 occur 3, 5, 4, 2 times. Find the mean and the median.

$n = 3 + 5 + 4 + 2 = 14$; $\sum fx = 3(1) + 5(2) + 4(3) + 2(4) = 33$, so the mean is $\frac{33}{14} \approx 2.36$.

Median: average of the 7th and 8th values. Running totals 3, 8, ... show both are in the score-2 group, so the median is 2.

Example 3 — Spread under transformations

Set $P = \{3, 5, 7, 9, 11\}$ has mean 7, range 8, and $\sigma = \sqrt{\frac{16+4+0+4+16}{5}} = \sqrt{8}$. Set Q is every value of P plus 10; set R is every value of P times 2.

Q : mean 17, range 8, $\sigma = \sqrt{8}$ (shift: spread unchanged).

R : mean 14, range 16, $\sigma = 2\sqrt{8}$ (scaling: spread doubles).

6 · PRACTICE SET — 37 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The mean of five numbers is 17. Four of the numbers are 12, 20, 15, and 25. What is the fifth number?

1. Total of all five: $5(17) = 85$.
2. Total of the four known numbers: $12 + 20 + 15 + 25 = 72$.
3. Fifth number = $85 - 72 = 13$.

Answer: 13

2. EASY · NO CALCULATOR NEEDED

What is the median of the data set 14, 9, 22, 17, 9, 31, 26, 12?

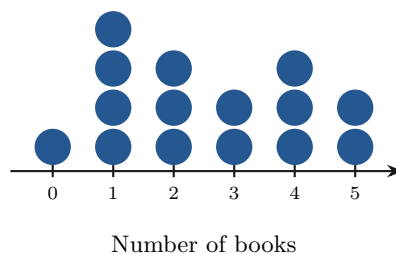
- (A) 9 **(B) 15.5 ✓**
- (C) 17 (D) 17.5

1. Order the 8 values: 9, 9, 12, 14, 17, 22, 26, 31.
2. With $n = 8$, the median is the average of the 4th and 5th values: $\frac{14 + 17}{2} = 15.5$.

Answer: (B)

3. EASY · NO CALCULATOR NEEDED

The dot plot shows the number of books each of 15 students read last month. What is the median number of books read?



- (A) 3 (B) 1
- (C) 2 ✓** (D) 2.5

1. There are $1 + 4 + 3 + 2 + 3 + 2 = 15$ dots, so the median is the 8th value in order.
2. Running totals: $0 \rightarrow 1$, $1 \rightarrow 5$, $2 \rightarrow 8$. The 8th value is 2.

Answer: (C)

4.

EASY · NO CALCULATOR NEEDED

A data set of 6 values has a mean of 20. A seventh value, 34, is added to the data set. What is the mean of the 7 values?

(A) 22 ✓

(B) 24

(C) 27

(D) 21

1. Original total: $6(20) = 120$.2. New total: $120 + 34 = 154$.3. New mean = $\frac{154}{7} = 22$.**Answer: (A)**

5.

EASY · CALCULATOR

The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

Pets	0	1	2	3
Students	7	11	8	4

1. Number of students: $7 + 11 + 8 + 4 = 30$.2. Total pets: $0(7) + 1(11) + 2(8) + 3(4) = 0 + 11 + 16 + 12 = 39$.3. Mean = $\frac{39}{30} = 1.3$.**Answer: 1.3**

6.

MEDIUM · NO CALCULATOR NEEDED

The table shows the scores of 30 students on a quiz. What is the median score?

Score	1	2	3	4	5
Frequency	5	6	4	9	6

(A) 4

(B) 3.2

(C) 3

(D) 3.5 ✓

1. $n = 30$, so the median is the average of the 15th and 16th scores.2. Running totals: score 1 $\rightarrow$ 5, score 2 $\rightarrow$ 11, score 3 $\rightarrow$ 15, score 4 $\rightarrow$ 24.3. The 15th score is 3 and the 16th score is 4, so the median is $\frac{3+4}{2} = 3.5$.**Answer: (D)**

7.

MEDIUM · NO CALCULATOR NEEDED

The mean of 7 test scores is 78. When one score is removed, the mean of the remaining 6 scores is 80. What was the score that was removed?

1. Total of 7 scores: $7(78) = 546$.
2. Total of the remaining 6: $6(80) = 480$.
3. Removed score = $546 - 480 = 66$.

Answer: 66

8.

MEDIUM · NO CALCULATOR NEEDED

Class A has 24 students with a mean test score of 82. Class B has 16 students with a mean test score of 72. What is the mean test score of all 40 students?

- (A) 80 (B) 77
(C) 78 ✓ (D) 79

1. Class A total: $24(82) = 1968$. Class B total: $16(72) = 1152$.
2. Combined total: $1968 + 1152 = 3120$.
3. Combined mean = $\frac{3120}{40} = 78$. (The simple average of 82 and 72 is 77, which ignores the class sizes.)

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

A data set has a mean of 40, a median of 38, a standard deviation of 6, and a range of 25. Every value in the data set is multiplied by 2, and then 3 is subtracted from each result. What is the standard deviation of the new data set?

- (A) 9 (B) 12 ✓
(C) 15 (D) 24

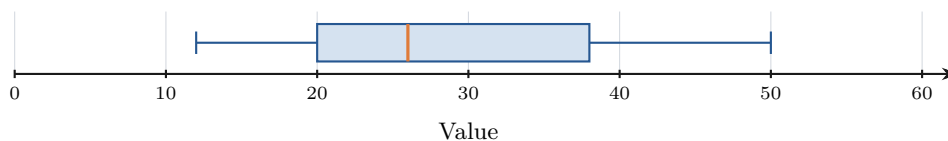
1. For $y = 2x - 3$, the standard deviation is $|2| \times 6 = 12$.
2. Subtracting 3 shifts every value equally, so it does not change the spread; only the multiplication by 2 matters.

Answer: (B)

10.

MEDIUM · NO CALCULATOR NEEDED

The box plot summarizes a data set. Which of the following must be true about the data set?



(A) The standard deviation of the data set is 18.

(B) Exactly half of the data values are between 12 and 26.

(C) The mean of the data set is 26.

(D) At least 25% of the data values are greater than or equal to 38. ✓

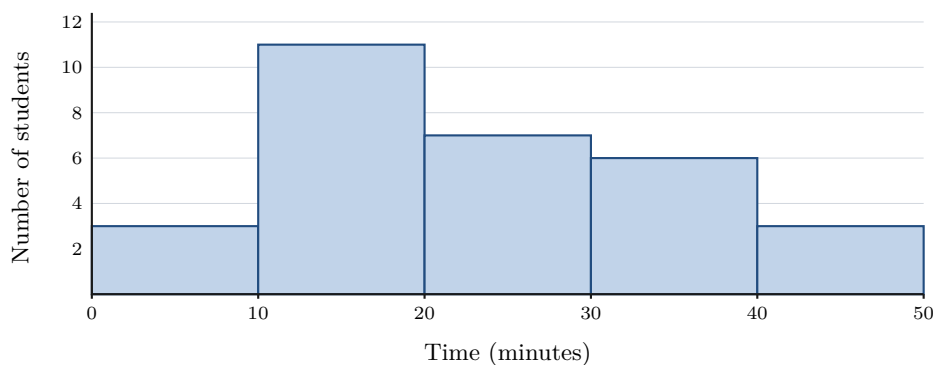
1. Reading the plot: min 12, $Q_1 = 20$, median 26, $Q_3 = 38$, max 50.
2. The upper whisker and upper half of the box together hold the top quarter of the data: at least 25% of the values are greater than or equal to $Q_3 = 38$. This must be true.
3. A box plot does not show the mean, so 26 (the median) cannot be assumed to be the mean; the IQR is $38 - 20 = 18$, which is not the standard deviation; and the median only splits the data into halves, so “exactly half” between 12 and 26 cannot be guaranteed.

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

The histogram shows the distribution of the times, in minutes, that 30 students took to finish a puzzle. In which interval does the median time lie?



(A) 20 to 30 ✓

(B) 30 to 40

(C) 0 to 10

(D) 10 to 20

1. Bar heights: 3, 11, 7, 6, 3 (total 30).
2. With $n = 30$, the median is the average of the 15th and 16th values.
3. Running totals: 0–10 → 3, 10–20 → 14, 20–30 → 21. Both the 15th and 16th values lie in the 20–30 bar. (The tallest bar, 10–20, is not the median bar.)

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

The data set 6, 9, 11, 14, 20 has a median of 11. A sixth value x is added to the data set, and the median of the new six-value data set is 12.5. Which of the following could be the value of x ?

(A) 13

(B) 16 ✓

(C) 8

(D) 12

1. With six values, the median is the average of the 3rd and 4th values. We need $\frac{3\text{rd} + 4\text{th}}{2} = 12.5$, i.e. $3\text{rd} = 11$ and $4\text{th} = 14$.
2. That happens only when $x \geq 14$ (then the ordered list is 6, 9, 11, 14, ...).
3. Check: $x = 16$ gives 6, 9, 11, 14, 16, 20, median 12.5. The values 8, 12, 13 give medians 10, 11.5, 12.

Answer: (B)**13.**

MEDIUM · NO CALCULATOR NEEDED

Maria's scores on her first five quizzes are 78, 91, 84, 88, and 80. What score must she earn on her sixth quiz so that the mean of all six scores is exactly 85?

1. Needed total for six quizzes: $6(85) = 510$.
2. Total so far: $78 + 91 + 84 + 88 + 80 = 421$.
3. Sixth score = $510 - 421 = 89$.

Answer: 89**14.**

HARD · NO CALCULATOR NEEDED

Group A has 30 members with a mean score of 70. Group B has a mean score of 90. The mean score of all the members of both groups combined is 78. How many members are in Group B?

1. Let b be the number in Group B. Total score: $70(30) + 90b$. Combined members: $30 + b$.
2. $\frac{2100 + 90b}{30 + b} = 78 \Rightarrow 2100 + 90b = 2340 + 78b$.
3. $12b = 240 \Rightarrow b = 20$. (Check: $\frac{2100 + 1800}{50} = 78$.)

Answer: 20

15.

HARD · NO CALCULATOR NEEDED

The mean of 12 values is 45. A new value is added, and the mean of all 13 values is 47. What is the value that was added?

1. Original total: $12(45) = 540$. New total: $13(47) = 611$.
2. Added value = $611 - 540 = 71$.

Answer: 71**16.**

HARD · NO CALCULATOR NEEDED

Data set A is 0, 10, 10, 10, 10, 10, 20. Data set B is 3, 3, 5, 10, 15, 17, 17. Both data sets have a mean of 10. Which of the following statements about the standard deviations of the two data sets is true?

- (A) Data set A has the greater standard deviation because more of its values equal the mean.
- (B) Data set A has the greater standard deviation because its range is greater.

(C) Data set B has the greater standard deviation because its values are generally farther from the mean. ✓

- (D) The two standard deviations are equal because the means are equal.

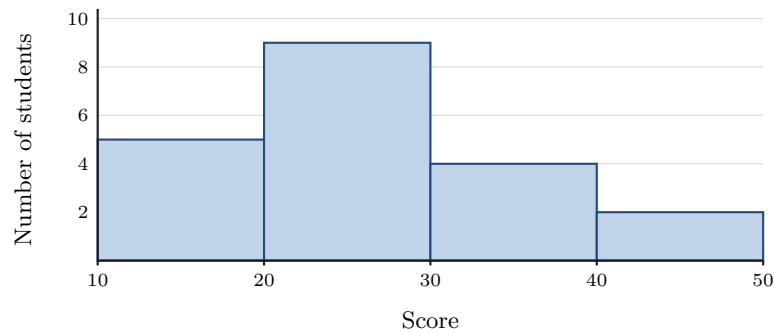
1. Standard deviation depends on how far the values are from the mean (10), not on the range.
2. Data set A : deviations $-10, 0, 0, 0, 0, 0, 10$; sum of squares = $100 + 100 = 200$.
3. Data set B : deviations $-7, -7, -5, 0, 5, 7, 7$; sum of squares = $49 + 49 + 25 + 0 + 25 + 49 + 49 = 246$.
4. Both have $n = 7$, so B has the greater standard deviation, even though A has the greater range (20 versus 14).

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

The histogram shows the distribution of scores for 20 students on a test. Each interval includes its left endpoint but not its right endpoint. Which of the following could be the mean score of the 20 students?



(A) 24 ✓

(B) 33

(C) 38

(D) 19

1. Bar heights: 5, 9, 4, 2 (total 20) on intervals 10–20, 20–30, 30–40, 40–50.

2. Least possible mean (every score at its interval's left endpoint): $\frac{5(10) + 9(20) + 4(30) + 2(40)}{20} = \frac{430}{20} = 21.5$.

3. Greatest possible mean (right endpoints, not quite reached): $\frac{5(20) + 9(30) + 4(40) + 2(50)}{20} = \frac{630}{20} = 31.5$.

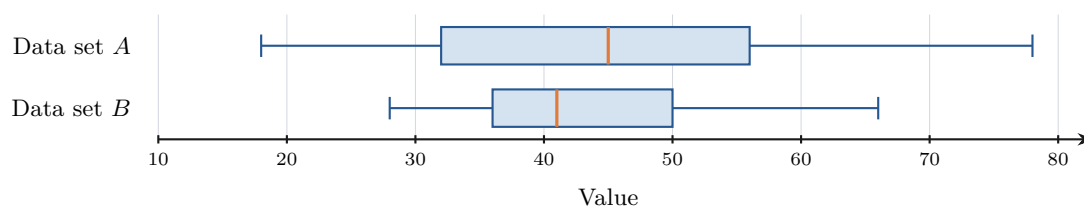
4. So the mean is at least 21.5 and less than 31.5. Among the choices only 24 is in that range.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

The box plots summarize two data sets, A and B . Which of the following statements must be true?



(A) More than 75% of the values in data set B are less than 50.

(B) The range of data set B is greater than the range of data set A .

(C) The mean of data set A is greater than the mean of data set B .

(D) At least 25% of the values in data set A are greater than 50. ✓

1. Data set A : min 18, $Q_1 = 32$, median 45, $Q_3 = 56$, max 78. Data set B : min 28, $Q_1 = 36$, median 41, $Q_3 = 50$, max 66.
2. At least 25% of the values in A are greater than or equal to $Q_3 = 56$, and $56 > 50$, so at least 25% of the values in A are greater than 50. This must be true.
3. The means cannot be read from box plots. About 75% of B is at most 50, but values equal to 50 may be included, so “more than 75% less than 50” is not guaranteed. The range of B is 38 and of A is 60, so B does not have the greater range.

Answer: (D)

19.

HARD · NO CALCULATOR NEEDED

The mean of the five numbers x , $x + 4$, $2x - 1$, $3x + 2$, and $x - 9$ is 20. What is the median of the five numbers?

1. Sum: $x + (x + 4) + (2x - 1) + (3x + 2) + (x - 9) = 8x - 4$.
2. Mean = $\frac{8x - 4}{5} = 20 \Rightarrow 8x - 4 = 100 \Rightarrow x = 13$.
3. The numbers are 13, 17, 25, 41, 4. In order: 4, 13, 17, 25, 41.
4. The median is the middle value, 17.

Answer: 17

20.

HARD · NO CALCULATOR NEEDED

Five distinct positive integers have a mean of 20 and a median of 18. What is the greatest possible value of the largest of the five integers?

1. Sum of the five integers: $5(20) = 100$. Order them $a < b < 18 < d < e$ (the median is the middle number, 18).
2. To make e as large as possible, make the others as small as possible: $a = 1$, $b = 2$, and $d = 19$ (the next integer above 18).
3. $e = 100 - 1 - 2 - 18 - 19 = 60$.

Answer: 60**21.**

HARD · NO CALCULATOR NEEDED

A data set of 9 values has a mean of 52 and a median of 50. The greatest value in the data set, 90, is replaced by 108. Which of the following describes the effect on the mean and median?

(A) The mean increases by 2 and the median increases by 18. (B) The mean and the median both increase by 2.

(C) The mean increases by 2 and the median is unchanged. ✓

(D) The mean increases by 18 and the median is unchanged.

1. The total increases by $108 - 90 = 18$, so the mean increases by $\frac{18}{9} = 2$.
 2. The median is the 5th value in order. The replaced value is the largest and stays the largest, so the median stays at 50.
- Answer:** (C)

22.

HARD · NO CALCULATOR NEEDED

The table shows the number of siblings of each of 15 students.

Siblings	0	1	2	3	4
Students	3	5	4	2	1

Four more students are surveyed, and each of them has more siblings than any of the original 15 students. What is the median number of siblings for all 19 students?

(A) 1.5

(B) 2 ✓

(C) 3

(D) 1

1. With 19 values, the median is the 10th value in order. The four new values are all larger than every original value, so they sit at the end and do not affect the first 15 positions.
2. Running totals of the original: $0 \rightarrow 3$, $1 \rightarrow 8$, $2 \rightarrow 12$. The 10th value is 2.
3. The median of the 19 students is 2 (the original median was 1).

Answer: (B)**23.**

HARD · NO CALCULATOR NEEDED

A data set has a mean of 30 and a standard deviation of 4. Each value x in the data set is replaced by $y = ax + b$, where $a > 0$. The new data set has a mean of 95 and a standard deviation of 10. What is the value of b ?

(A) 65

(B) 75

(C) 6

(D) 20 ✓

1. New standard deviation: $a \times 4 = 10 \Rightarrow a = 2.5$.
2. New mean: $a(30) + b = 95 \Rightarrow 75 + b = 95$.
3. $b = 20$.

Answer: (D)

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

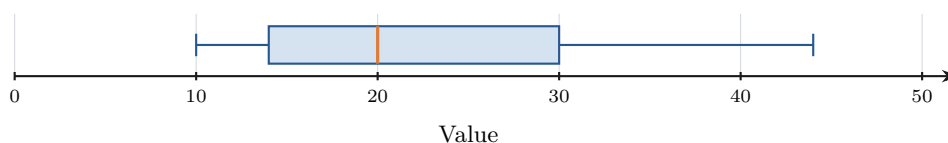
The data set 2, 6, k , 10, $3k$ consists of five numbers, where k is a positive integer. The mean of the data set equals its median. What is the value of k ?

1. The mean is $\frac{2+6+k+10+3k}{5} = \frac{18+4k}{5}$. The median depends on where k falls in the order, so split into cases.
2. $k \geq 11$: ordered list 2, 6, 10, k , $3k$, median 10. Mean = 10 $\Rightarrow 18 + 4k = 50 \Rightarrow k = 8$, which contradicts $k \geq 11$. No solution.
3. $6 \leq k \leq 10$: ordered list 2, 6, k , 10, $3k$, median k . Mean = $k \Rightarrow 18 + 4k = 5k \Rightarrow k = 18$, which contradicts $k \leq 10$. No solution.
4. $k = 4$ or 5: ordered list 2, k , 6, 10, $3k$, median 6. Mean = 6 $\Rightarrow 18 + 4k = 30 \Rightarrow k = 3$, which contradicts $k \in \{4, 5\}$. No solution.
5. $k = 2$ or 3: ordered list 2, k , 6, $3k$, 10 (for $k = 2$: 2, 2, 6, 6, 10), median 6. Mean = 6 $\Rightarrow k = 3$. Check: 2, 3, 6, 9, 10 has mean $\frac{30}{5} = 6$ and median 6. This works.
6. $k = 1$: list 1, 2, 3, 6, 10 has median 3 and mean 4.4. No solution.
7. The only value is $k = 3$.

Answer: 3**25.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The box plot summarizes a data set of 11 distinct whole numbers. For these 11 values, Q_1 is the median of the lowest five values and Q_3 is the median of the highest five values. What is the greatest possible sum of the 11 values?



1. In order $x_1 < x_2 < \dots < x_{11}$: the median is $x_6 = 20$, $Q_1 = x_3 = 14$, $Q_3 = x_9 = 30$, $\min x_1 = 10$, $\max x_{11} = 44$.
2. Make the free values as large as possible while keeping them distinct and in order: $x_2 \leq 13$; $x_4, x_5 \leq 18, 19$ (below 20); $x_7, x_8 \leq 28, 29$ (below 30); $x_{10} \leq 43$.
3. Sum = $10 + 13 + 14 + 18 + 19 + 20 + 28 + 29 + 30 + 43 + 44 = 268$.

Answer: 268

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

Five distinct positive integers have a mean of 10 and a median of 9. Which of the following could *not* be the greatest of the five integers?

(A) 13 ✓

(B) 17

(C) 25

(D) 28

1. Order the integers $a < b < 9 < d < e$ with sum $5(10) = 50$, so $a + b + d + e = 41$.
2. Greatest possible e : take $a = 1$, $b = 2$, $d = 10$, giving $e = 41 - 13 = 28$.
3. Least possible e : take a and b as large as possible, $a = 7$, $b = 8$; then $d + e = 26$ with $d < e$, so $e \geq 14$ (for example $d = 12$, $e = 14$).
4. So the greatest integer can be any value from 14 to 28. Of the choices, 13 is outside this range, so it cannot be the greatest.

Answer: (A)

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

Data set A is 2, 4, 6, 8, 10. Data set B consists of the values of A together with one additional value, 6. Data set C consists of the values of A together with the two additional values 0 and 12. Data set D consists of the values of A together with the two additional values 5 and 7. Consider the following statements about standard deviations.

- I. The standard deviation of B is less than that of A .
- II. The standard deviation of C is greater than that of A .
- III. The standard deviation of D is greater than that of B .

Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. All four data sets have mean 6. Variance $= \frac{\sum(x-6)^2}{n}$. For A : $\frac{16+4+0+4+16}{5} = 8$.
2. B adds a value at the mean (0 deviation): $\frac{40}{6} = \frac{20}{3} \approx 6.67 < 8$. I is true.
3. C adds deviations -6 and 6 : $\frac{40+72}{7} = 16 > 8$. II is true.
4. D adds deviations -1 and 1 : $\frac{40+2}{7} = 6$, which is less than $\frac{20}{3}$. So the standard deviation of D is *less* than that of B ; III is false.
5. Statements I and II only.

Answer: (B)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

Group A has a mean score of 72 and Group B has a mean score of 84. The mean score of all 45 people in the two groups combined is 80. A third group, C , of 15 people is then added, and the mean score of all 60 people in the three groups is 79. What is the mean score of Group C ?

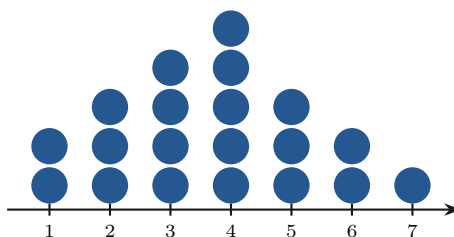
1. Total score of A and B : $45(80) = 3600$.
2. Total score of all three groups: $60(79) = 4740$.
3. Total score of Group C : $4740 - 3600 = 1140$.
4. Mean of Group $C = \frac{1140}{15} = 76$. (The sizes of A and B are not needed.)

Answer: 76

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The dot plot shows the number of books read last month by each of 20 students. Then k more students are surveyed, and each of them read 8 books. For how many positive integer values of k is the median of all $20 + k$ values still equal to the median of the original 20 values, while the mean of all $20 + k$ values is at least 4.5?



Number of books

1. Original: $n = 20$, total books $1(2) + 2(3) + 3(4) + 4(5) + 5(3) + 6(2) + 7(1) = 74$. The 10th and 11th values are both 4 (running totals 2, 5, 9, 14), so the median is 4.
2. Median condition: the value 4 occupies positions 10 through 14, and the new values go to the end. With $N = 20 + k$ values the median positions must stay within 10–14: for N even need $\frac{N}{2} + 1 \leq 14$ ($N \leq 26$); for N odd need $\frac{N+1}{2} \leq 14$ ($N \leq 27$). So $k \leq 7$.
3. Mean condition: $\frac{74 + 8k}{20 + k} \geq 4.5 \Rightarrow 74 + 8k \geq 90 + 4.5k \Rightarrow 3.5k \geq 16 \Rightarrow k \geq 4.57$, so $k \geq 5$.
4. Both conditions: $k = 5, 6, 7$. That is 3 values.

Answer: 3

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

The table shows the scores of 30 students on a quiz; the frequencies of scores 2 and 4 are a and b . The mean score is 3.2. What is the median score of the 30 students?

Score	1	2	3	4	5
Frequency	4	a	7	b	3

1. Total students: $4 + a + 7 + b + 3 = 30 \Rightarrow a + b = 16$.
2. Total of scores: $3.2(30) = 96$, and $1(4) + 2a + 3(7) + 4b + 5(3) = 40 + 2a + 4b$. So $2a + 4b = 56 \Rightarrow a + 2b = 28$.
3. Subtract $a + b = 16$: $b = 12$, so $a = 4$.
4. Frequencies 4, 4, 7, 12, 3; running totals 4, 8, 15, 27, 30. The median is the average of the 15th and 16th values: the 15th is 3 and the 16th is 4.
5. Median = $\frac{3+4}{2} = 3.5$.

Answer: 3.5

31.

SAT CHALLENGE • NO CALCULATOR NEEDED

A data set has eight values: 12, 15, 18, 20, 20, 22, 25, 28. The values 12 and 28 are removed and the values 19 and 21 are added in their place. Which of the following describes the mean and the standard deviation (SD) of the new data set compared with the original data set?

- (A) Mean smaller, SD smaller (B) Mean same, SD larger
- (C) Mean same, SD smaller ✓** (D) Mean same, SD same

1. Total removed: $12 + 28 = 40$. Total added: $19 + 21 = 40$. The sum is still 160, so the mean is still 20 (unchanged).
2. The two extreme values, 8 below and 8 above the mean, were replaced by values only 1 from the mean. With the mean fixed, the data are more tightly clustered, so the standard deviation is smaller. (Check: original SD ≈ 4.27 , new SD ≈ 2.87 .)
3. Mean unchanged, standard deviation smaller.

Answer: (C)

32.

SAT CHALLENGE · CALCULATOR

A data set consists of the five values 5, 8, 17, x , x , where x is a real number. The variance of a data set is the mean of the squared deviations of its values from their mean. What is the least possible value of the variance of this data set?

(A) 15.6 ✓

(B) 10

(C) 26

(D) 78

$$1. \sum v = 30 + 2x \text{ and } \sum v^2 = 378 + 2x^2, \text{ so } \sigma^2 = \frac{378 + 2x^2}{5} - \left(\frac{30 + 2x}{5} \right)^2.$$

$$2. \text{ Expand: } \sigma^2 = \frac{1890 + 10x^2 - 900 - 120x - 4x^2}{25} = \frac{6x^2 - 120x + 990}{25}.$$

$$3. \text{ This quadratic opens upward; its vertex is at } x = \frac{120}{12} = 10, \text{ the mean of } 5, 8, 17.$$

$$4. \sigma_{\min}^2 = \frac{600 - 1200 + 990}{25} = \frac{390}{25} = 15.6. \text{ (Check: mean} = 10, \frac{25 + 4 + 49 + 0 + 0}{5} = 15.6.)$$

Answer: (A)

33.

SAT CHALLENGE · CALCULATOR

A data set has 10 values: 31, 38, 42, 45, 47, 52, 53, 58, 65, 69. The mean of the data set is 50. The least value and the greatest value are removed, leaving 8 values. Which of the following is true of the new data set compared with the original? (SD means standard deviation.)

(A) Mean unchanged; SD unchanged

(B) Mean smaller; SD smaller

(C) Mean greater; SD smaller

(D) Mean unchanged; SD smaller ✓

1. The removed values sum to $31 + 69 = 100$, so they average 50, exactly the mean. Removing them leaves the mean unchanged: the remaining sum is $500 - 100 = 400$ and $\frac{400}{8} = 50$.

2. The two removed values are the farthest from the mean (19 below and 19 above), so every deviation that remains is smaller than 19 in size. The sum of squared deviations drops sharply while 8 values remain, so the standard deviation decreases.

3. (Numerically, the population standard deviation falls from about 11.3 to about 8.2.)

Answer: (D)

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

A list of nine positive integers has a mean of 12 and a median of 9. What is the greatest possible value of the sum of the three largest integers in the list?

1. Mean $\times$ count: the nine integers sum to $9 \cdot 12 = 108$.
2. Order them $x_1 \leq x_2 \leq \dots \leq x_9$. The median is the middle value, so $x_5 = 9$. The three largest are x_7, x_8, x_9 .
3. To make $x_7 + x_8 + x_9$ as large as possible, make the other six values as small as possible. x_1, x_2, x_3, x_4 can each be 1 (positive integers). $x_5 = 9$ is fixed. $x_6 \geq x_5 = 9$, so the smallest choice is 9.
4. Sum of the other six: $4(1) + 9 + 9 = 22$.
5. Greatest possible sum of the three largest: $108 - 22 = 86$. (Attainable: 1, 1, 1, 1, 9, 9, 28, 29, 29 has sum 108, median 9, and top three summing to 86.)

Answer: 86**35.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Nine positive integers, not necessarily distinct, have a sum of 100 and a median of 9. What is the greatest possible value of the mean of the four largest of these integers?

(A) 25

(B) 20.25

(C) 21.75 ✓

(D) 23.75

1. Order them $a_1 \leq a_2 \leq \dots \leq a_9$. The median is $a_5 = 9$, and $a_6, \dots, a_9 \geq 9$.
2. To make the top four as large as possible, make $a_1, \dots, a_4$ as small as possible. Repeats are allowed, so each can equal 1.
3. Then $a_6 + a_7 + a_8 + a_9 = 100 - (1 + 1 + 1 + 1) - 9 = 87$ (for example 9, 9, 9, 60, which respects the ordering).
4. Mean = $\frac{87}{4} = 21.75$.
5. (Forcing distinct values 1, 2, 3, 4 would give $\frac{81}{4} = 20.25$.)

Answer: (C)

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has six values: 2, 3, 5, 12, 18, 20. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.

- I. The mean is 4 greater.
- II. The standard deviation is smaller.
- III. The median is 4 greater.

Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II and III

1. Original: sum = 60, mean = 10, median = $\frac{5+12}{2} = 8.5$. After adding 4 to each value: mean 14, median 12.5, values 6, 7, 9, 16, 22, 24.
2. I: the appended value is the new mean, 14, so the mean stays 14, which is 4 more than 10. True.
3. II: adding 4 to each value does not change the sum of squared deviations (306). Appending the mean adds 0 to it, while the number of values grows from 6 to 7, so the standard deviation falls. True.
4. III: the seven values in order are 6, 7, 9, 14, 16, 22, 24, so the median is 14, not $8.5 + 4 = 12.5$. False.
5. Statements I and II only.

Answer: (B)

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has a median of 52 and an interquartile range (IQR) of 14. Each value x in the data set is replaced by $15 - \frac{x}{2}$. What is the sum of the median and the IQR of the new data set?

(A) 3

(B) 11

(C) -18

(D) -4 ✓

1. Median transforms like every value: $15 - \frac{52}{2} = -11$.
2. Spread (IQR) is multiplied by the absolute value of the multiplier and is not changed by the added constant: $\left| -\frac{1}{2} \right| \cdot 14 = 7$.
3. Sum: $-11 + 7 = -4$.
4. (Check with the data 30, 44, 46, 50, 54, 58, 60, 70: median 52, IQR $59 - 45 = 14$; new values give median -11, quartiles -14.5 and -7.5, IQR 7.)

Answer: (D)